Test 3 Wed. 55 min.
covers HW's 10, 11, 12, 13

Will have the opportunity to
do problems on take-home exam
and have scores averaged (as in Test 2).

I should have exams graded by Friday class.

One week of classes after Fall break
2 HW's due ~~Wed~~ Thurs. Dec. 5th

Sol'ns to HW's 12 + 13 posted

Recall

$u_{xx} - u_{tt} = 0 \qquad 0 \le x < \infty$

$u(x,0) = \phi(x) \in C^2$

$u_t(x,0) = \psi(x) \in C^1$

$u(0,t) = f(t)$ Dirichlet

$\Delta u = 0$

$n=1 \quad u_{xx} = 0$

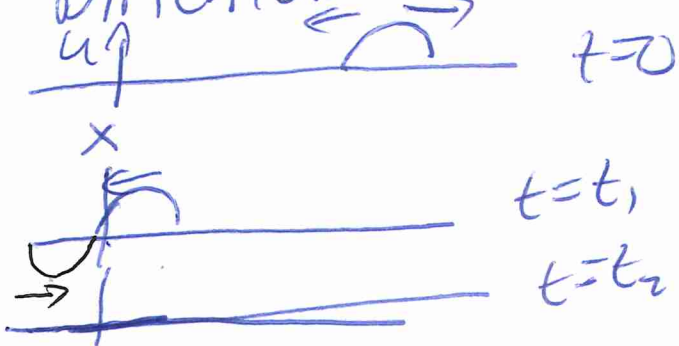
$u = c_1 x + c_2$

2 conditions c_1 & c_2

$\cancel{\phi(x) = 0}$

$f(t) = 0$ ✓

Dirichlet extend oddly



Initial boundary value problem.

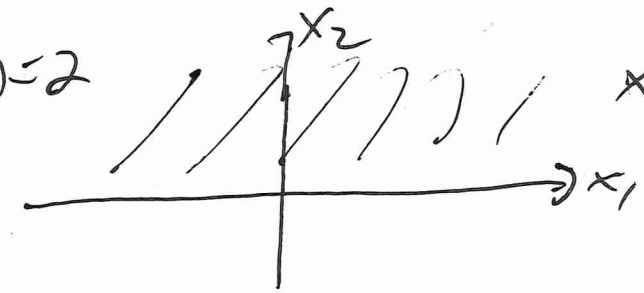
Aside $\Delta u = 0$

$\Delta u - u_{tt} = 0$

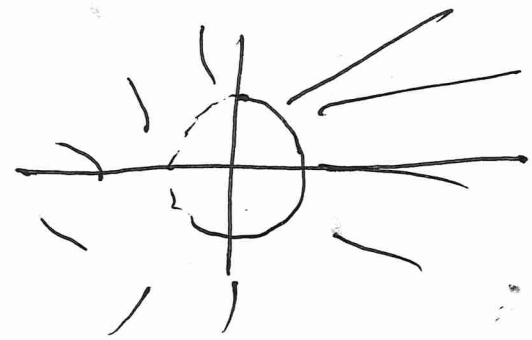
semi-infinite domain

$n=1 \quad u(0,t) = f(t)$ well-posed

$n=2$



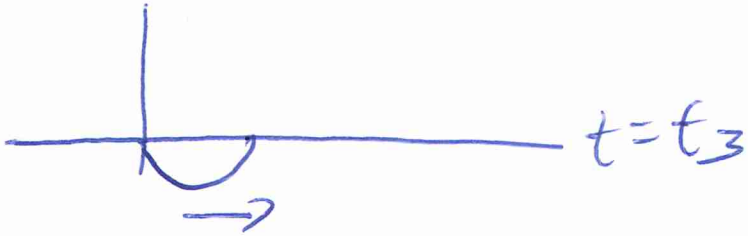
$x_2 > 0$



$|\underline{x}| > a$

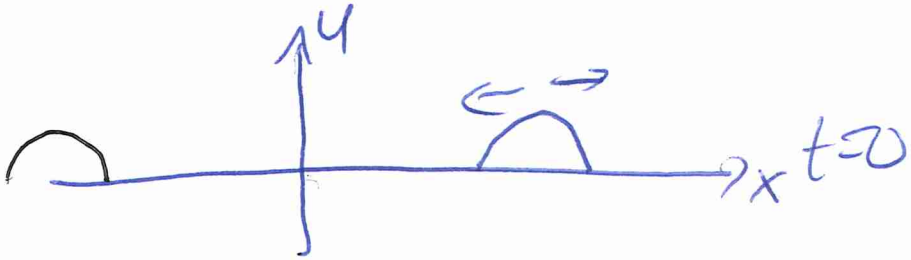
$\lim_{\underline{x} \to \infty} u$ bounded well-posed

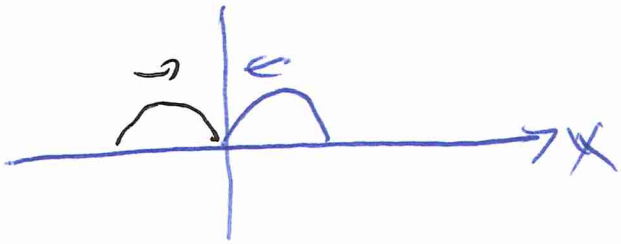
$n=3 \quad \lim_{\underline{x} \to \infty} u = 0$ well-posed

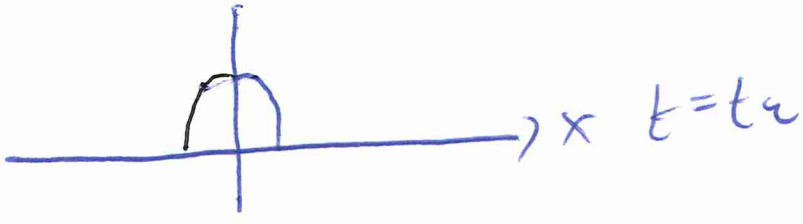


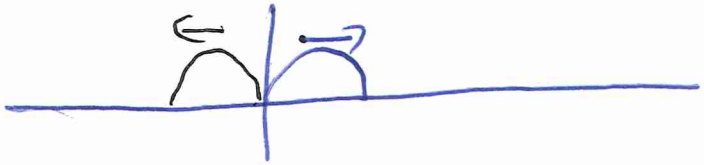
$\frac{\partial u}{\partial x}(0,t) = 0$ Neumann BC.

extend data evenly







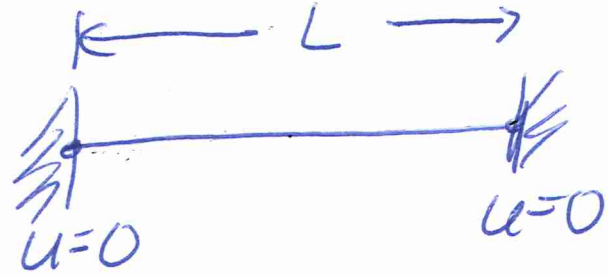


~~Spring~~

Slinky experiment!

Ch 8 S8 Vibrating String

Going from one BC to two



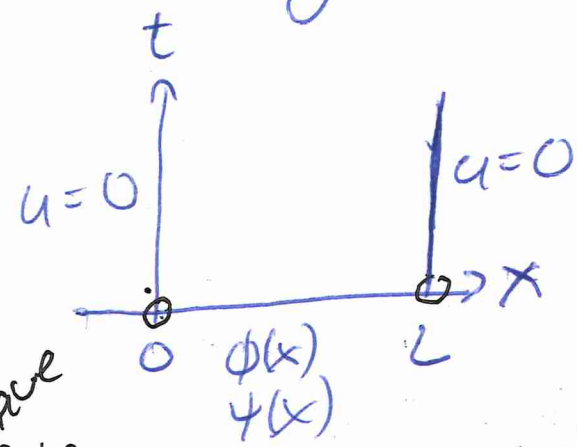
guitar string.

$u_{xx} - u_{tt} = 0 \qquad 0 < x < L, \; t > 0$

$u(x,0) = \phi(x) \qquad 0 < x < L$

$u_t(x,0) = \psi(x) \qquad 0 < x < L$

$u(0,t) = u(L,t) = 0 \qquad t > 0$



don't have to have for wave eq

$$\begin{cases} \phi(x=0,0) = u(x=0,t=0) = 0 \\ \phi(x=L,0) = u(x=L,t=0) = 0 \end{cases}$$

compatibility condition

Sol'n separation of Variables

1st consider $u_{xx} - u_{tt} = 0$

$u(0,t) = u(L,t) = 0$

Assume $u(x,t) = X(x)\,T(t)$

subst. in

$$\left[X''T - XT'' = 0 \right] \frac{1}{XT}$$

separate

$$\frac{X''}{X} - \frac{T''}{T} = 0 \Rightarrow \frac{X''}{X} = \frac{T''}{T} = -\lambda$$

Space $X'' + \lambda X = 0 \quad 0 < x < L$

BC $u(0,t) = X(0)\,T(t) = 0$

$X(0) = 0 \qquad X(L) = 0$

Case 1 $\lambda < 0$ assume e^{rx}

$$X(x) = c_1 e^{\sqrt{-\lambda}x} + c_2 e^{-\sqrt{-\lambda}x}$$

$$\left.\begin{aligned} X(0) &= c_1 + c_2 = 0 \\ X(L) &= c_1 e^{\sqrt{-\lambda}L} + c_2 e^{-\sqrt{-\lambda}L} = 0 \end{aligned}\right\} \quad c_1 = c_2 = 0$$

$$A\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\det A \neq 0$

$X(x) = 0 \Rightarrow u(x,t) = 0$

trivial soln : uninteresting.

<u>Case 2</u> $\lambda = 0$ $X'' = 0$

$X'(x) = c_1$ $X(x) = c_1 x + c_2$

$X(0) = c_2 = 0$ $X(x) = c_1 x$ $X(L) = c_1 L = 0$

$\Rightarrow c_1 = 0$

$X(x) = 0$ trivial soln :(

Case 3 $\lambda > 0$

$$X'' + \lambda X = 0$$

$$X(x) = c_1 \cos(\sqrt{\lambda}\, x) + c_2 \sin(\sqrt{\lambda}\, x)$$

$$X(0) = c_1 (1) + c_2 (0) = c_1 = 0$$

$$X(x) = c_2 \sin(\sqrt{\lambda}\, x)$$

$$X(L) = c_2 \sin(\sqrt{\lambda}\, L) = 0$$

$$\sqrt{\lambda}\, L = k\pi \qquad k = 0, \pm 1, \pm 2, \ldots$$

$$\lambda_k = \frac{k^2 \pi^2}{L^2} \qquad k = 1, 2, 3, \ldots$$

<u>eigenvalues</u> for this problem

$$X(x) = \sin\left(\frac{k\pi}{L} x\right)$$

$T(t)$: $T'' + \lambda_k T = 0 \qquad \lambda_k > 0$

$$T_k(t) = A_k \cos\left(\frac{k\pi}{L} t\right) + B_k \sin\left(\frac{k\pi}{L} t\right)$$

No IC here!!

$$u_k(x,t) = X_k(x) T_k(t) = \sin\left(\frac{k\pi x}{L}\right)\left[A_k \cos\left(\frac{k\pi}{L} t\right) + B_k \sin\left(\frac{k\pi}{L} t\right)\right]$$

Now apply I.C.

$$u_k(x,0) = A_k \sin\left(\frac{k\pi x}{L}\right) = \phi(x)$$ Fourier sine series!

$$\frac{\partial u_k}{\partial t}(x,0) = \underbrace{\frac{k\pi}{L} B_k}_{\text{coeff of Fourier sine series of } \psi(x).} \sin\left(\frac{k\pi}{L} x\right) = \psi(x)$$

$$u(x,t) = \sum_{k=1}^{\infty} \sin\left(\frac{k\pi x}{L}\right)\left[A_k \cos\left(\frac{k\pi}{L} t\right) + B_k \sin\left(\frac{k\pi}{L} t\right)\right]$$

$x=0 \quad u=0$ ✓ $\qquad x=L \quad u=0$ ✓